

Deep Learning for Image Analysis

Midterm Project 2025

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I. Data Preparation & Annotation

We used the *PCB Defects Dataset* from Kaggle ([link](#)) containing printed circuit board (PCB) images exhibiting four defect types: missing hole, mouse bite, open circuit, and short circuit.

Annotation was conducted using Roboflow, where defect regions were labeled using bounding boxes. Labels were exported in YOLO format. A total of 1,957 annotated samples were created.

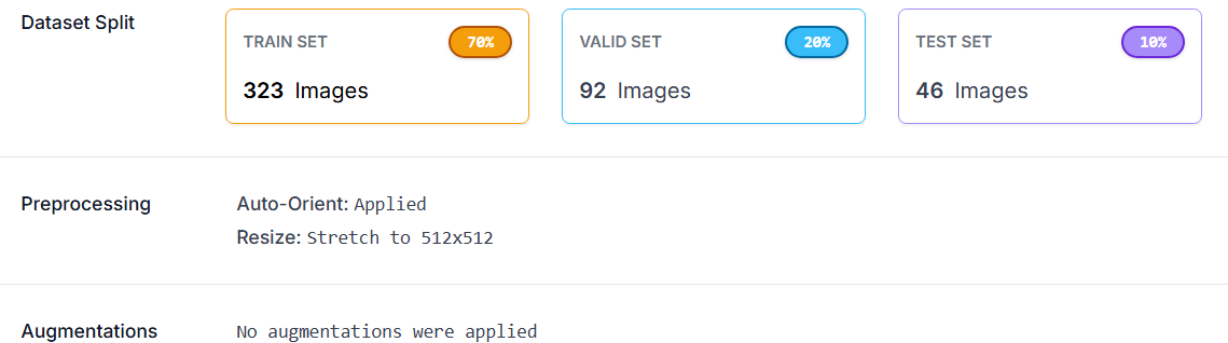


Fig.1 Baseline Dataset Preparation

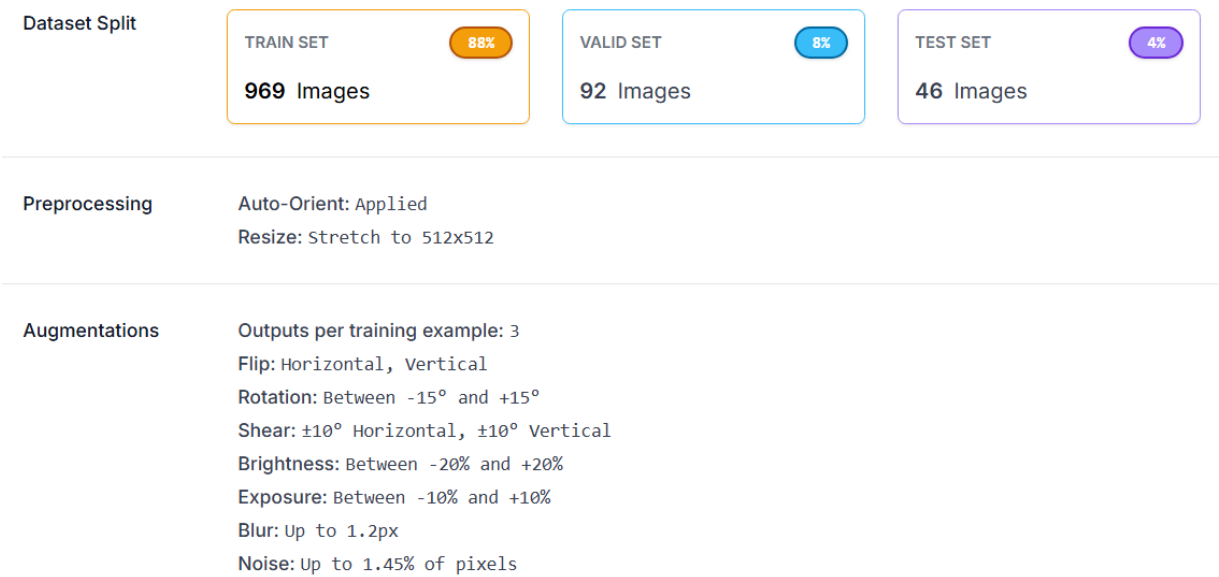


Fig.2 Enhanced Dataset Preparation

COLOR ?	CLASS NAME	COUNT ↻
●	missing_hole	492
●	mouse_bite	492
●	open_circuit	482
●	short_circuit	491

Fig.3 Class Distribution

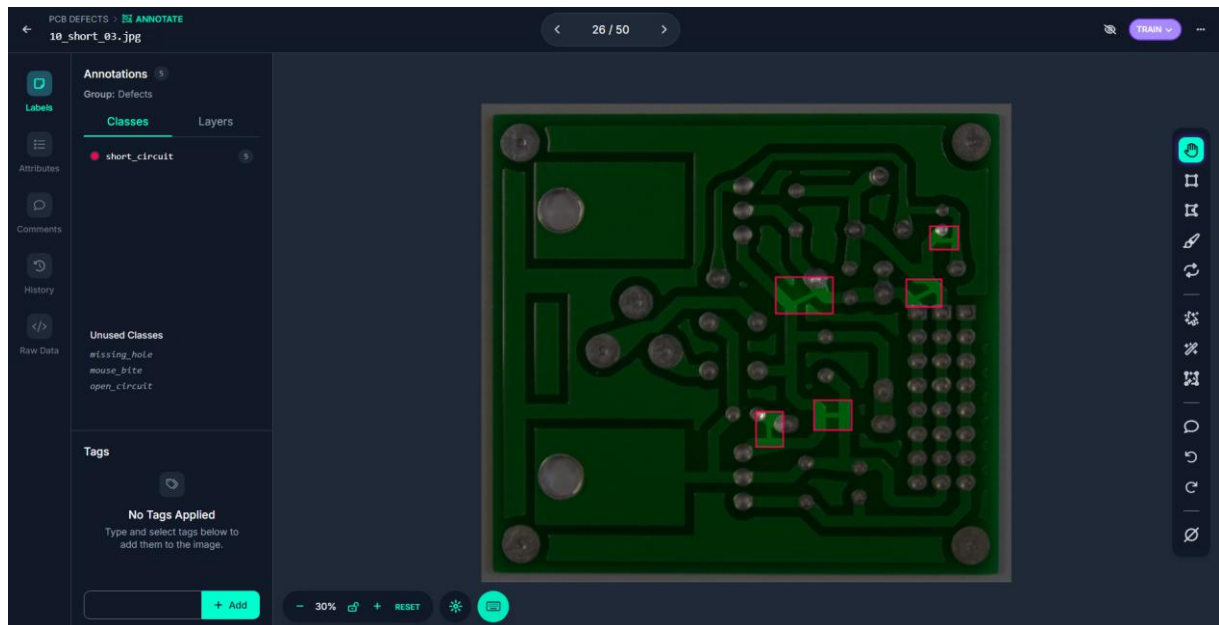


Fig.4 Labeling Interface

II. Model Development

We trained a YOLOv12 object detection model for PCB defect identification.

Two experimental pipelines were implemented for comparison:

Pipeline	Description
Baseline	Trained directly on raw data without any preprocessing or augmentation.
Enhanced	Trained using preprocessing and augmentation techniques to improve robustness.

1. Environment:

Python 3.10.19, Torch 2.6.0+cu124, Ultralytics 8.3.227, RTX 4060 GPU.

2. Data Augmentation Summary:

Method	Purpose
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Rotation ($\pm 15^\circ$)	Improves angle robustness.
90° Flip (CW/CCW)	Increases orientation invariance.
Shear ($\pm 10^\circ$)	Models' geometric distortion.
Brightness ($\pm 20\%$)	Adapts to lighting variation.
Exposure ($\pm 10\%$)	Adapts to lighting variation.
Blur ($\leq 1.2\text{px}$)	Handles motion blur.
Noise ($\leq 1.45\%$)	Improves noise tolerance.

3. Why YOLOv12

Chosen for its lightweight design, real-time inference capability, and enhanced *Flush Attention* mechanism improving small-defect detection compared to YOLOv8.

4. Model Comparison and Discussion

Both models used the same YOLOv12 architecture and dataset.

The Baseline Model trained on raw images, while the Enhanced Model used augmentation (rotation, zoom, shear, blur, noise, color jitter).

Metric	Baseline	Enhanced	Better
Recall	0.851	0.871	Enhanced
F1-score	0.890	0.92	Enhanced
mAP@0.5	0.918	0.925	Enhanced
mAP@0.5–0.95	0.420	0.478	Enhanced

The Enhanced Model performs better overall.

Augmentation improved its robustness to lighting, angle, and noise changes, helping it detects subtle defects such as open circuits that the baseline often missed.

III. Evaluation

Both models were evaluated on the same test set using Precision, Recall, F1-score, and mAP@0.5.

1. Precision

Both models reached perfect precision, but at different confidence levels. The Baseline Model achieved 1.00 precision at 0.697, while the Enhanced Model reached it at 0.803. The higher confidence point of the Enhanced Model shows it makes more reliable predictions with fewer false positives. Its precision curve is smoother and remains high across all classes, indicating better overall stability and accuracy.

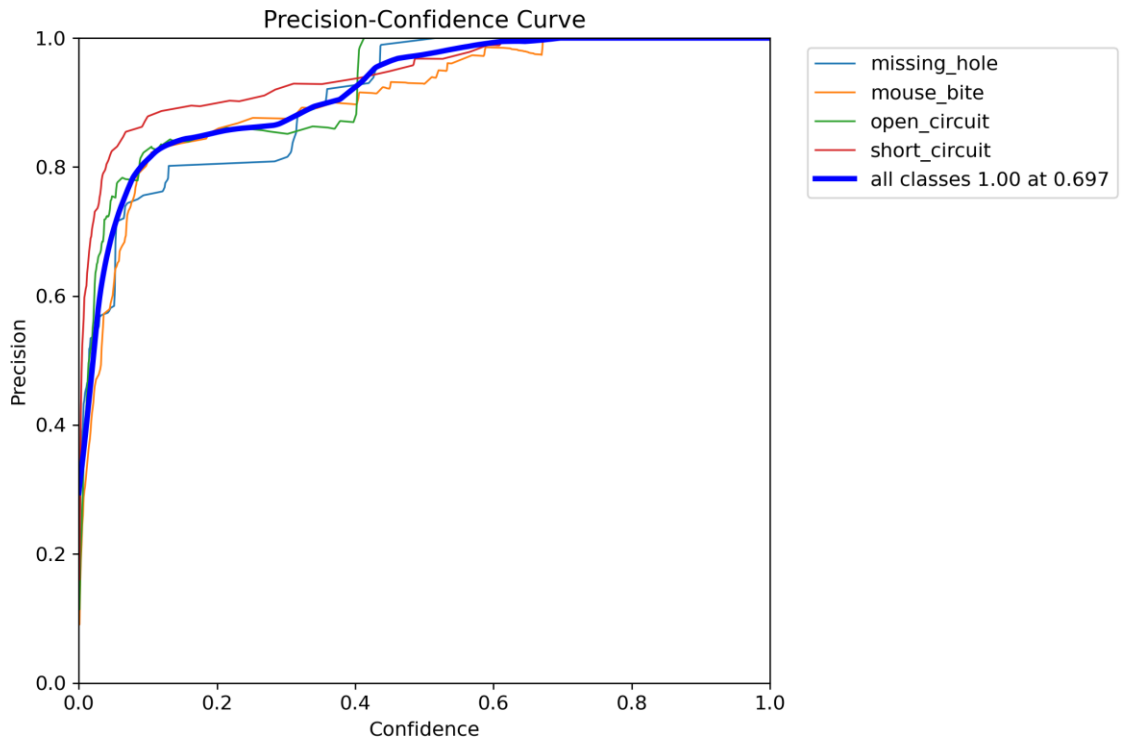


Fig.5 Baseline Precision

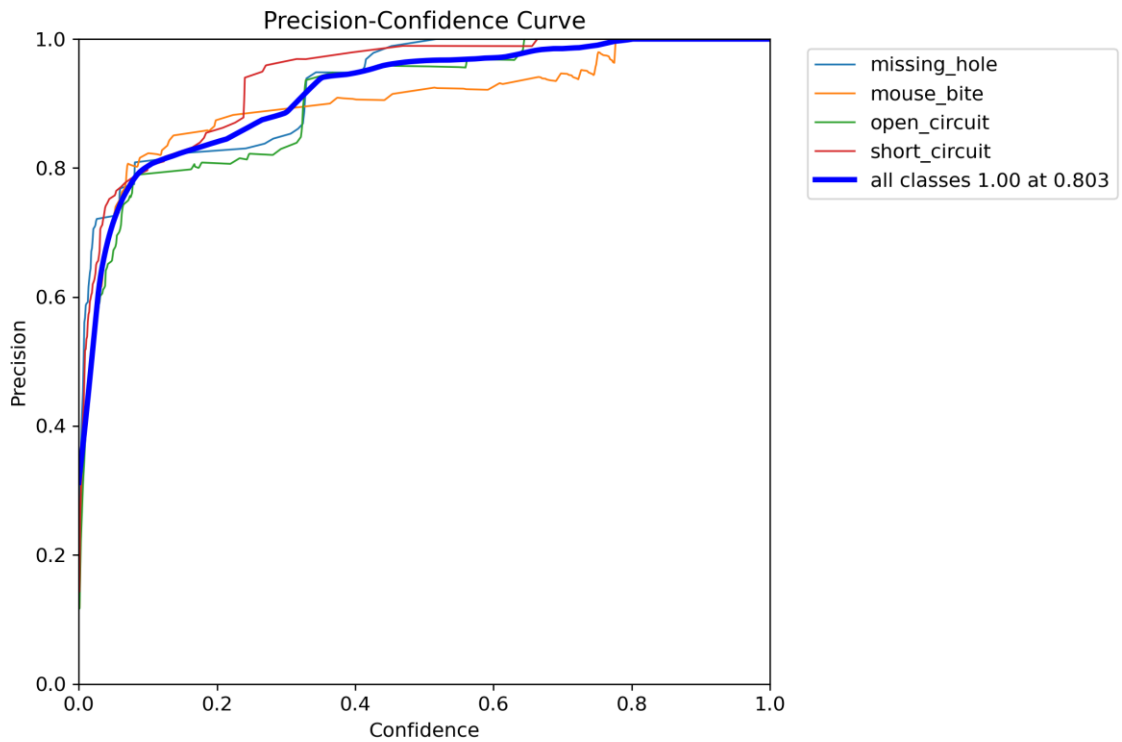


Fig.6 Enhanced Precision

2. Recall

Both models achieved a maximum recall of 0.96, but their confidence behaviors differ. The Enhanced Model maintains higher recall across most confidence ranges, especially

between 0.2 and 0.7, where it detects more true defects while keeping stability across all classes. The Baseline Model drops more sharply as confidence increases, indicating less consistency at higher thresholds. Overall, the Enhanced Model provides better recall balance and stronger detection reliability, especially for small or subtle defects.

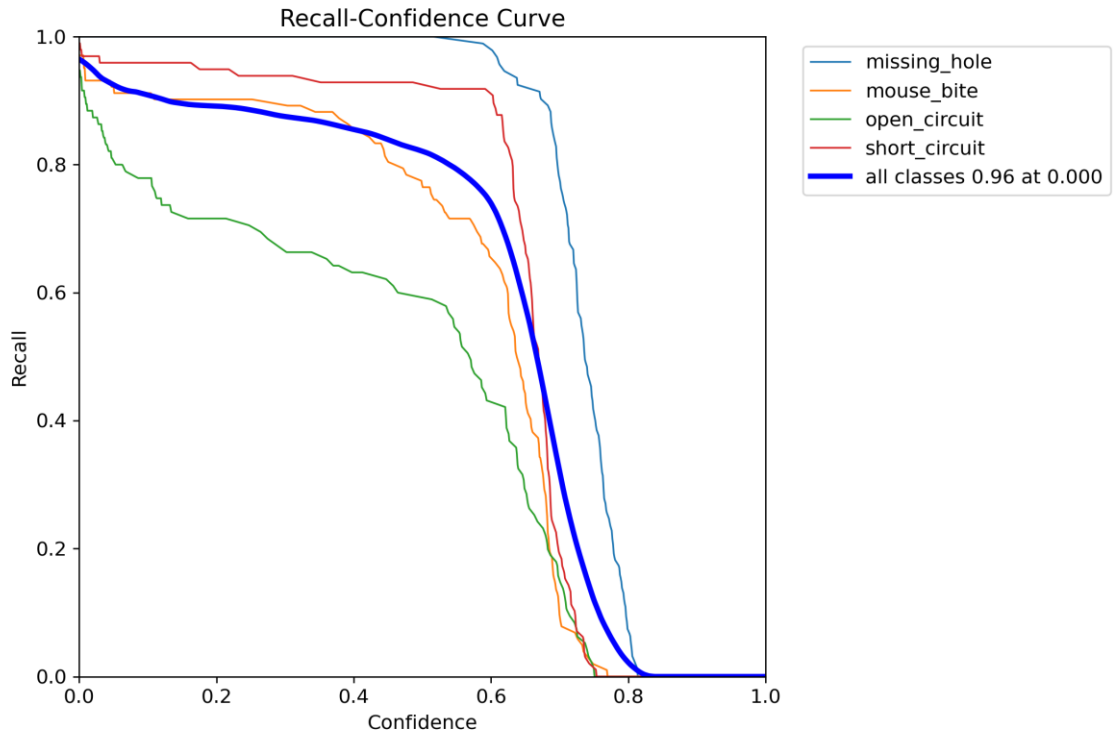


Fig.7 Baseline Recall

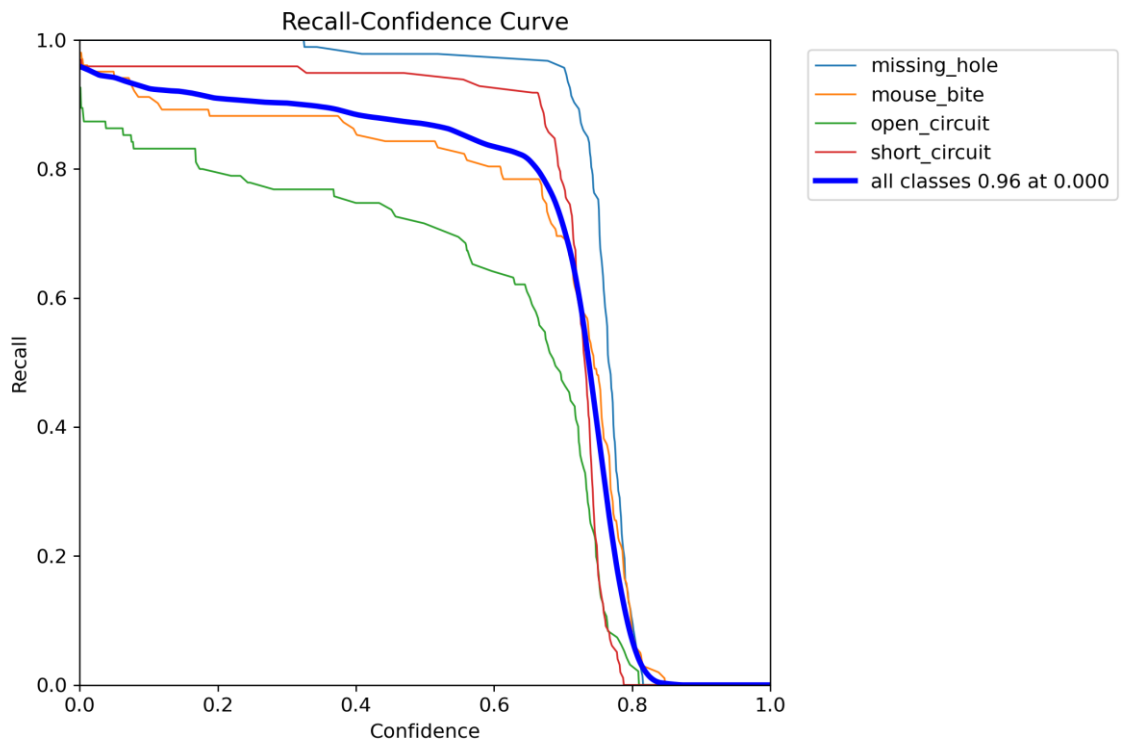


Fig.8 Enhanced Recall

3. F1-score

The Baseline Model achieved a maximum F1 score of 0.89 at a confidence of 0.435, while the Enhanced Model reached 0.92 at 0.357. The Enhanced Model's curve is consistently higher and smoother, showing stronger balance between precision and recall across all classes. Indicating improved overall detection consistency and robustness.

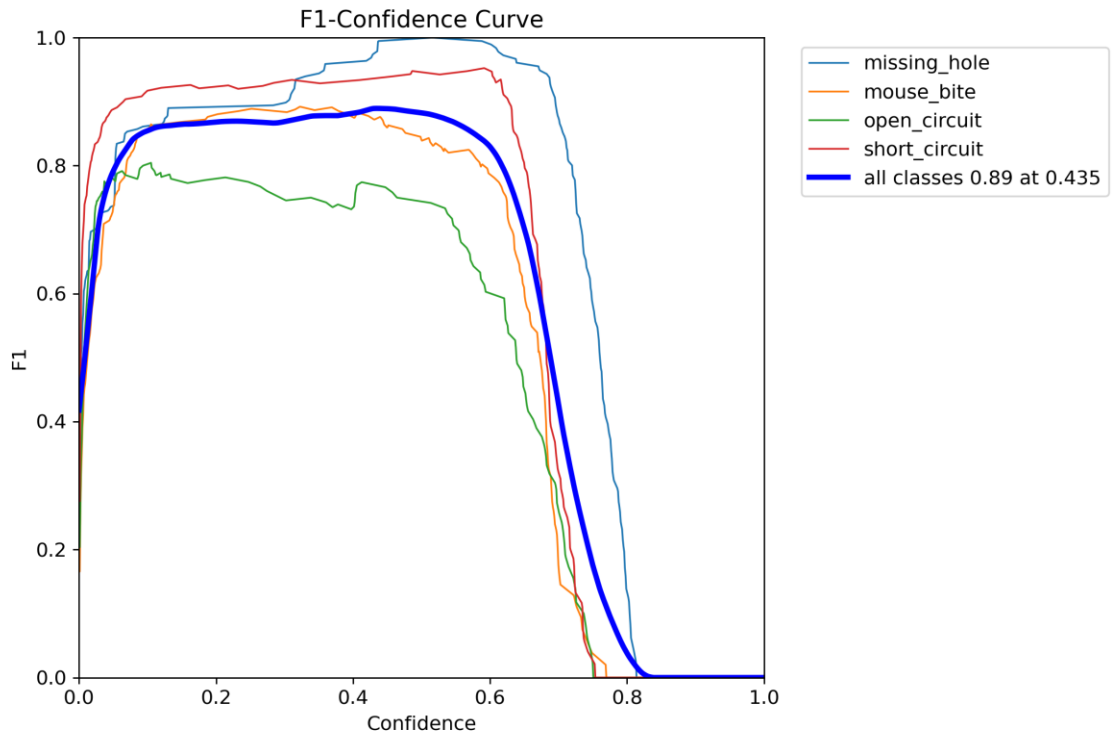


Fig.9 Baseline F1-score

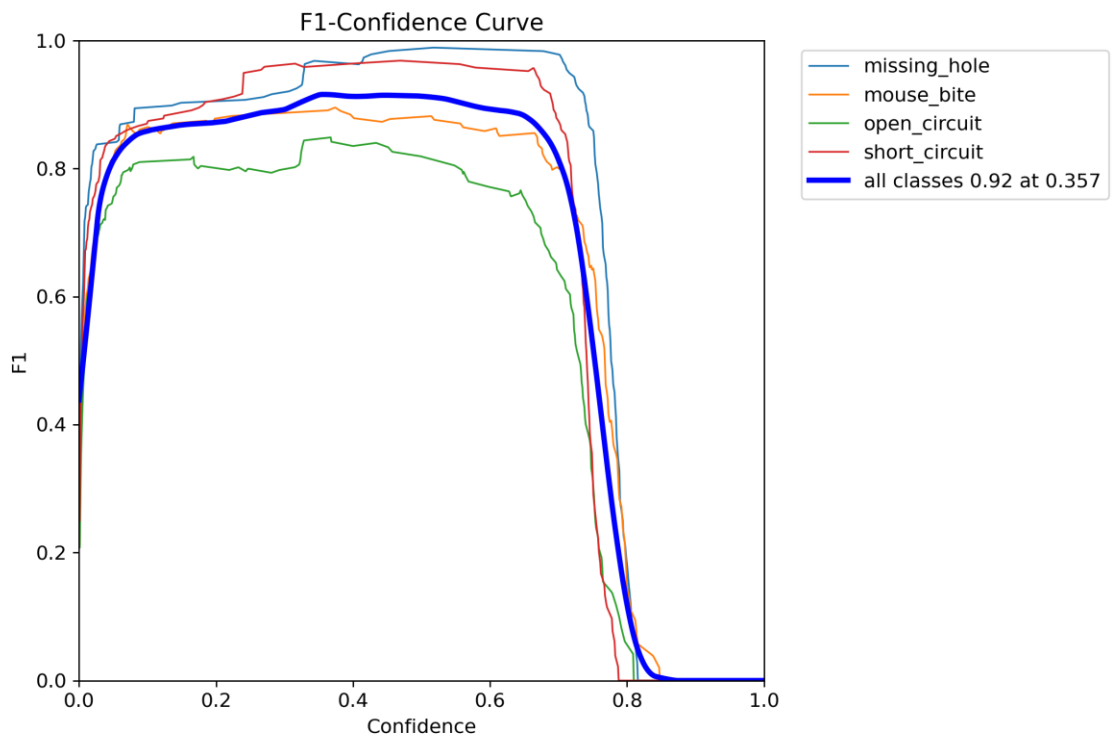


Fig.10 Enhanced F1-score

4. mAP@0.5

The Enhanced Model reached higher mAP@0.5 values and converged faster than the Baseline Model. It surpassed 0.80 before Epoch 20, while Baseline required about 40 epochs to reach the same level. Throughout training, the Enhanced Model's curve is smoother and remains above Baseline, indicating better learning stability and stronger generalization. By the final epochs, both models approach similar peaks, but the Enhanced Model maintains consistently higher precision-recall balance across all stages.

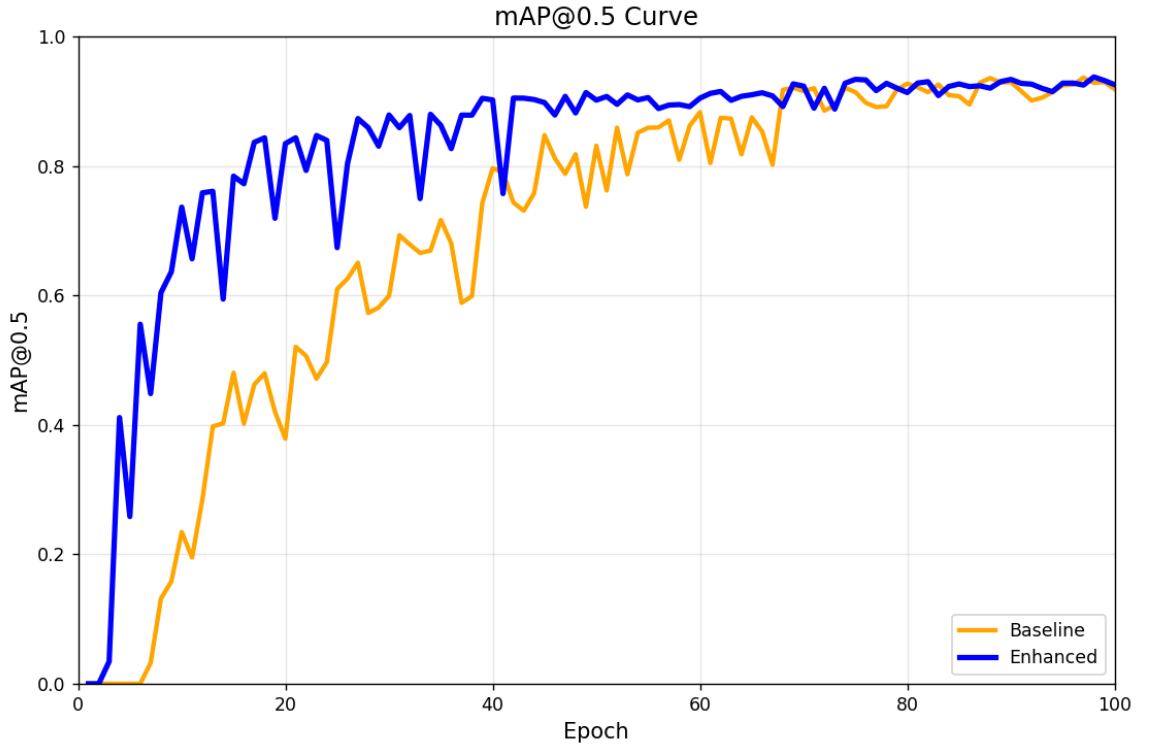


Fig.11 mAP@0.5 Baseline vs. Enhanced

5. Qualitative Results

Both models show a steady decrease in loss and improvement in precision, recall, and mAP over epochs. The Enhanced Model exhibits smoother and faster convergence across all loss terms (box, cls, dfl) and validation metrics compared to the Baseline Model.

In particular:

- Validation losses for the Enhanced Model are consistently lower, indicating better generalization.
- Precision and Recall curves rise more rapidly and stabilize earlier.
- mAP@0.5 and mAP@0.5–0.95 grow faster and reach higher final values.

Overall, the Enhanced Model demonstrates clearer learning stability, reduced fluctuation, and stronger performance across all evaluation metrics.

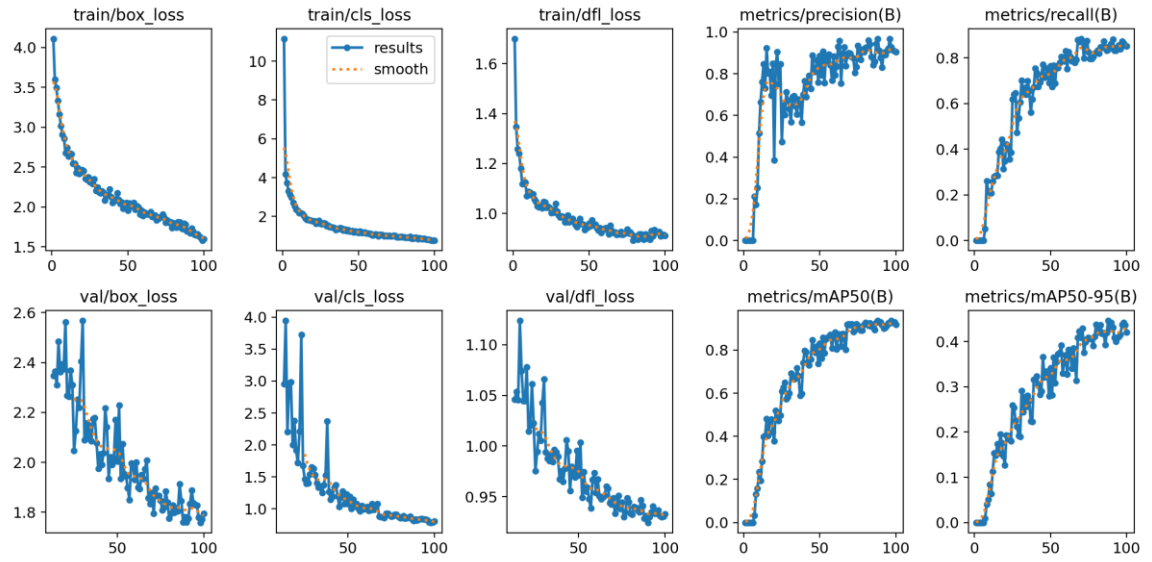


Fig.12 Baseline Result

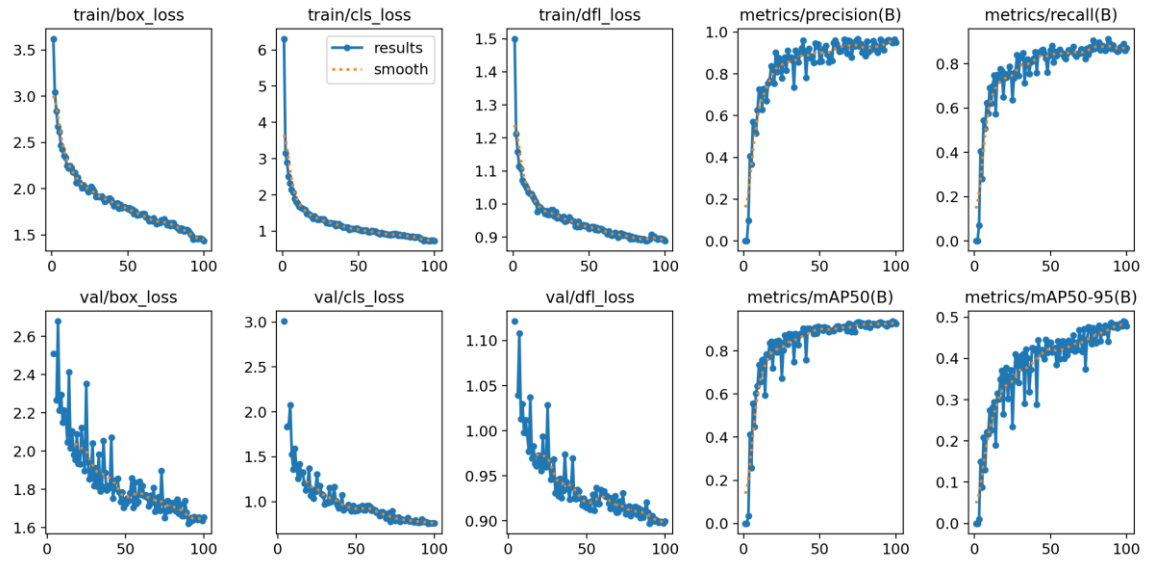


Fig.13 Enhanced Result

6. Sample Prediction

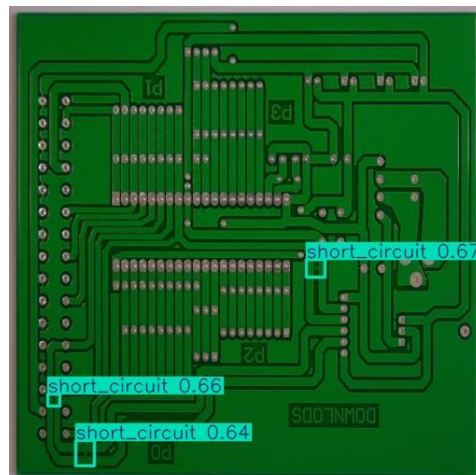


Fig.14 Baseline Sample Prediction

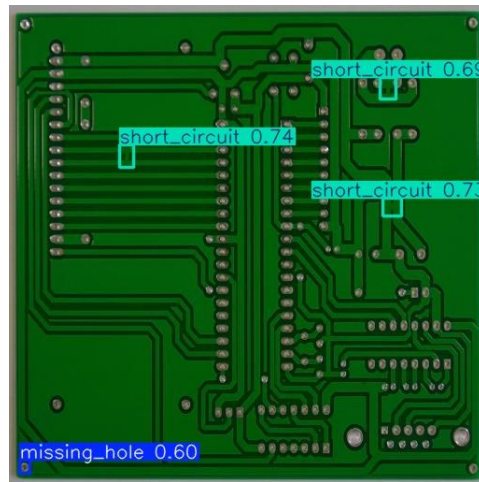


Fig.15 Enhanced Sample Prediction

IV. Analysis & Discussion

1. Open circuit is a major weakness of Baseline

(1) Evidence

In the F1-Score graph, the baseline's open circuit F1 peak score is only about 0.80, far lower than the Enhanced Model's 0.95.

(2) Possible reasons

An "open circuit" is a tiny gap or break in a circuit trace. Visually, it is a very subtle defect that can easily be confused with the normal trace background.

(3) How Enhanced Model solved

The Enhanced Model uses data augmentation. This forces the model to learn subtle features more carefully, explaining why the Enhanced Model achieves a significant improvement in its F1 score on open circuits.

2. False Positives: Confusion caused by low confidence thresholds

(1) Evidence

(2) Possible reasons

The baseline model may be unable to distinguish between real defects and visual interference. It may make silkscreen lines, solder reflections, or uneven lighting for mouse bite or short circuit, resulting in a large number of low-confidence false alarms.

(3) How Enhanced Model solved

Enhanced models utilize data augmentation to simulate various visual disturbances in the test environment. This forces the model to learn more carefully to distinguish between "real features" and "noise," thus significantly reducing false positives and achieving better operational confidence.

V. Model architecture diagram

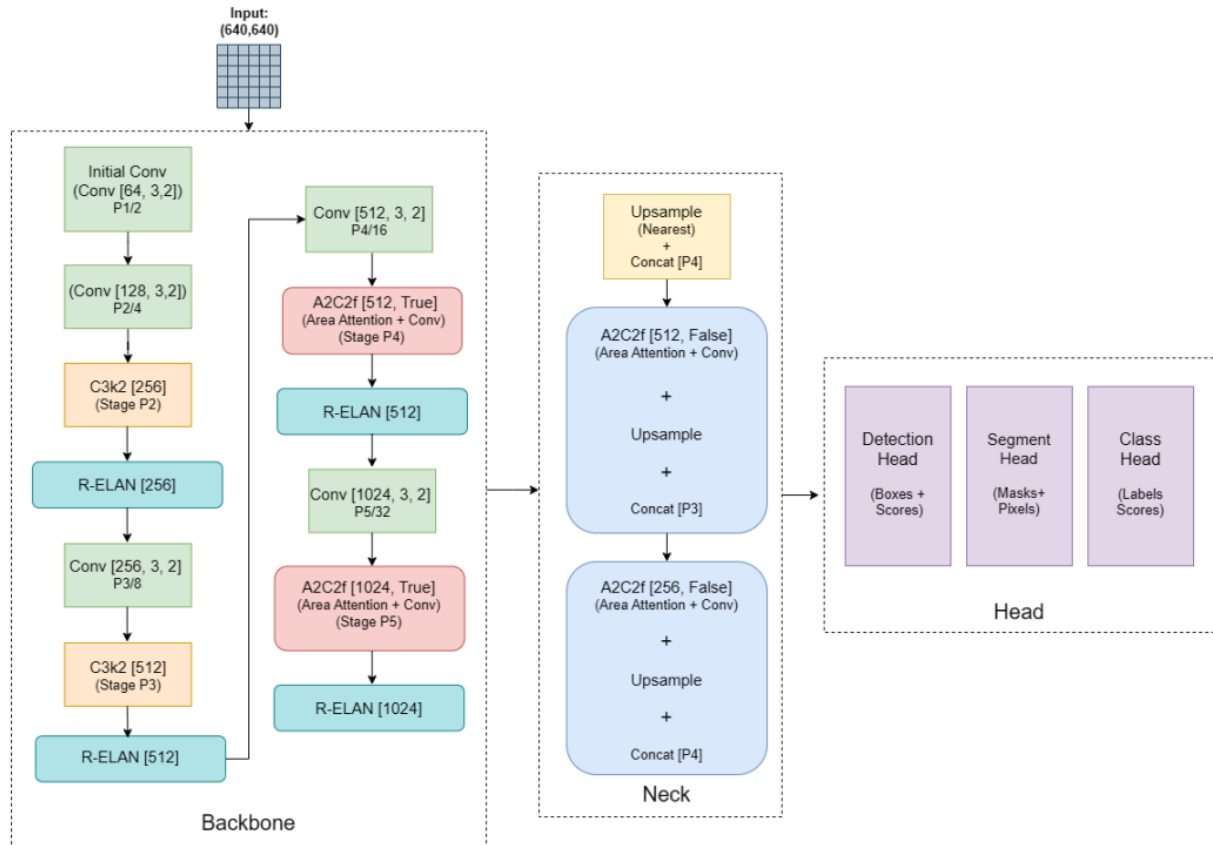


Fig.16 YOLOv12 Model architecture

1. Backbone
 - Purpose: Feature Extractor
 - Function: This is the model's "eyes." It receives a 640x640 PCB image as input and then extracts key visual features (such as edges, textures, and shapes) layer by layer through a series of Conv and R-ELAN modules.
2. Neck
 - Purpose: Feature Fusion
 - Function: The Neck's job is to fuse deep and shallow features. It uses Unsample and Concat operations to combine deep and shallow features, allowing the model to detect both large and small defects simultaneously.
3. Head
 - Purpose: Predictor.
 - Function: Function: This is the model's "brain" and "mouth." It receives feature maps fused from the Neck at three different scales and uses the Flush Attention module to output the detection results.